

AI Literature Review and Synthesis in Research

Topic overview

AI-supported literature reviewing spans scholarly discovery, study selection, information extraction, evidence appraisal, and the construction of written syntheses. This report uses a decomposition framework covering mechanisms, applications, limitations, and comparisons across five stages of that work. The evidence is comparatively mature for prioritizing screening and increasingly persuasive for some verified extraction workflows. Retrieval-grounded synthesis has advanced rapidly, while comprehensive coverage, interpretive depth, and reliable evaluation remain unresolved. The landscape includes education, psychology, information systems, software engineering, materials science, ecology, qualitative methodology, and biomedicine; validation in humanities and many non-English contexts remains thin in the retrieved evidence. [Bolaños et al. \(2024\)](#) [Campos et al. \(2024\)](#) [Gartlehner et al. \(2025\)](#) [Asai et al. \(2026\)](#) [Foley et al. \(2025\)](#)

Executive synthesis

The central conclusion is task-specific augmentation. AI can reduce effort and improve some outputs when the task, source collection, evaluation criteria, and reviewer responsibilities are explicit. A system that finds useful papers, extracts a variable correctly, or writes a fluent paragraph has demonstrated different capabilities. None of those accomplishments alone establishes that an entire literature review is comprehensive, critically reasoned, or reproducible. [Wagner et al. \(2022\)](#) [Agarwal et al. \(2025\)](#)

Screening has the clearest historical continuity. Earlier text-mining research already distinguished prioritizing what a person reads from automatically excluding studies. Contemporary educational and software-engineering results reinforce the need to measure missed relevant studies and evaluate stopping rules within the review at hand. [O'Mara-Eves et al. \(2015\)](#) [Campos et al. \(2024\)](#) [Huotala et al. \(2025\)](#)

Synthesis depends on more than citation existence. Fabricated references are one failure mode; real references can still support inaccurate statements, conceal disagreement, or omit entire lines of inquiry. Retrieval-backed systems address source access, but long-form benchmarks continue to expose weaknesses in coverage and verifiability. [Walters and Wilder \(2023\)](#) [Li et al. \(2025\)](#) [Patel et al. \(2025\)](#)

Human oversight is a workflow to evaluate. Verification can improve an assisted process, but adding a human does not guarantee superior performance. The important comparison includes output quality and the total effort of prompting, reading, checking, correcting, and documenting decisions. [Gartlehner et al. \(2025\)](#) [Vaccaro et al. \(2024\)](#)

Interpretive reviewing requires additional standards. Qualitative and conceptual reviews must preserve context, reflexivity, disagreement, and theoretical relationships. Their success cannot be reduced to matching a reference list or producing a plausible theme label. [Foley et al. \(2025\)](#) [Pizard et al. \(2025\)](#) [Messerli and Crockett \(2024\)](#)

Scope and evidence boundaries

This is a focused narrative evidence map, not an exhaustive systematic review or a pooled meta-analysis. The search comprised one reconnaissance query followed by 20 sequential Consensus queries. The 20 searches returned 381 record appearances and 318 distinct Consensus IDs; including reconnaissance yielded 401 appearances and 329 distinct IDs before publication-version consolidation. These are retrieval counts, not counts of eligible studies. Thirty-seven paper records were fetched for closer assessment; 34 are cited in this report.

The synthesis draws principally on returned abstracts and paper records, with selective publisher, proceedings, and arXiv checks for publication status, metadata, affiliations, and methodological detail. It does not represent full-text appraisal of all 34 papers. All cited research papers were returned by Consensus during this session. Linked titles and in-text research citations use their complete canonical Consensus URLs. Supplemental metadata checks concern those same papers and are documented in the appendix.

Search 2 largely retrieved reviews about AI in education rather than evaluations of AI-assisted reviewing. Search 18 returned just one off-topic record and supplied no directly usable evidence. Several other broad queries also produced irrelevant material. These results were not used to infer effectiveness or disciplinary coverage. Dedicated review-oriented searches were completed, but no cross-disciplinary meta-analysis of the complete AI literature-review workflow was established.

Sub-area	Mechanisms and applications	Main evaluation question
1 Finding and mapping	Semantic retrieval, query expansion, citation networks	Which relevant traditions and studies remain undiscovered?
2 Selecting and organizing	Active learning, ranking, eligibility decisions, clustering	How much work is saved at an acceptable level of recall?
3 Reading and extracting	Structured extraction, comparison tables, appraisal support	Are values, relationships, and judgments faithful to their sources?
4 Synthesizing and writing	Multi-document synthesis, retrieval-grounded generation, themes	Are claims supported, disagreements retained, and coverage adequate?
5 Human AI workflows	Task allocation, verification, transparency, evaluation	Does the complete workflow improve quality and total effort?

Start here and priority reading order

This seven-paper sequence begins with an orientation review, moves through foundational approaches, and then examines current capabilities and a central conceptual critique. Publication prestige and citation counts are navigation aids, not substitutes for evaluating methods.

1 Broad orientation

Artificial intelligence for literature reviews: opportunities and challenges. Bolaños et al., 2024; 233 citations. Maps established review tools and the transition to LLM-based assistance. While reading: Distinguish the feature inventory from direct comparative evidence of effectiveness.

2 Historical foundation

Using text mining for study identification in systematic reviews: a systematic review of current approaches.

O'Mara-Eves et al., 2015; 685 citations. Explains why prioritization, automatic exclusion, and workload reduction require different judgments. While reading: Read how recall losses accompany some reported efficiency gains.

3 Foundation beyond biomedicine

Artificial intelligence and the conduct of literature reviews. Wagner et al., 2022; 259 citations. *REPEAT HIT*

across 3 sub-areas. Places AI-assisted reviews within information systems and social-science research methods. While reading: Treat its research agenda as a conceptual contribution rather than an outcome trial.

4 Current synthesis frontier

Synthesizing scientific literature with retrieval-augmented language models. Asai et al., 2026; 78 citations.

REPEAT HIT across 4 sub-areas. Evaluates retrieval-grounded scientific answers across four research domains.

While reading: Separate its multi-paper question-answering benchmark from validation of a complete literature review.

5 Current extraction frontier

arXiv2Table: Toward Realistic Benchmarking and Evaluation for LLM-Based Literature-Review Table

Generation. Wang et al., 2026; 8 citations. *ACL 2026 conference paper; index year 2025*. Tests literature-review tables with noisy retrieval and without supplying the desired schema. While reading: Inspect whether cell correctness also preserves comparisons between papers.

6 Current evaluation frontier

DeepScholar-Bench: A Live Benchmark and Automated Evaluation for Generative Research Synthesis. Patel et

al., 2025; 32 citations. *arXiv preprint; peer-reviewed version not established*. Measures research synthesis, retrieval, and verifiability together in a live benchmark. While reading: The reported composite metric is not a percentage of correct facts; this remains preprint evidence.

7 Key conceptual controversy

Artificial intelligence and illusions of understanding in scientific research. Messeri and Crockett, 2024; 624

citations. Explains how AI use could narrow scientific understanding and viewpoints. While reading: Evaluate the conceptual mechanisms and proposed risks separately from empirical estimates of error.

How the field got here

The field predates chatbots. Early work used text mining to reduce the effort of identifying studies, linking information retrieval and evidence synthesis. Active learning later made reviewer feedback central to deciding what to read next. Generative models expanded the ambition from selecting and extracting to composing related-work sections and answering questions across papers; recent benchmarks now test those broader claims. Ananiadou et al. (2009) de la Torre-López et al. (2023) van de Schoot et al. (2021) Agarwal et al. (2025)

Year	Milestone	Significance
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Year	Milestone	Significance
2009	Text mining supports systematic reviews Ananiadou et al. (2009)	Automation was already a collaborative methods problem.
2015	Review of study-identification automation O’Mara-Eves et al. (2015)	Recall and workload tradeoffs become explicit.
2021	ASReview framework van de Schoot et al. (2021)	Open active-learning workflows connect human labels and screening order.
2022	Information-systems research agenda Wagner et al. (2022)	AI reviewing is framed beyond clinical review production; online first in 2021.
2023	Multidisciplinary citation audit Walters and Wilder (2023)	Fluent reviews reveal fabricated references and errors in real citations.
2024	DiscipLink and LitSearch Zheng et al. (2024) Ajith et al. (2024)	Discovery gains user-centered and benchmark-based evaluation.
2025	Related-work and long-form benchmarks Agarwal et al. (2025) Li et al. (2025) Patel et al. (2025)	Retrieval and synthesis are assessed together, with new evaluation limitations.
2026	OpenScholar and arXiv2Table publications Asai et al. (2026) Wang et al. (2026)	Citation-grounded answers and relationally accurate tables become concrete test targets.

Terminology to retain when searching older work: text mining, automatic text classification, technology-assisted review, screening prioritization, relevance feedback, and semi-automation. Contemporary additions include dense retrieval, retrieval-augmented generation or RAG, scientific question answering, automated survey generation, generative research synthesis, and deep-research agents. These terms partly overlap; they do not describe one interchangeable technology. [de la Torre-López et al. \(2023\)](#) [Zhang et al. \(2023\)](#) [Patel et al. \(2025\)](#)

Sub-area guides

1 Finding and mapping literature

AI supports discovery through query expansion, semantic retrieval, recommendation, and visual organization. DiscipLink evaluates an interactive approach to crossing disciplinary vocabularies, while LitSearch shows that retrieval performance can differ substantially by system on realistic scientific queries. Both support assisted discovery, but neither demonstrates exhaustive retrieval across all fields. [Zheng et al. \(2024\)](#) [Ajith et al. \(2024\)](#)

The main distinction is between retrieving useful items and covering the relevant literature. LitSearch contains 597 queries centered on machine learning and natural-language processing; its results cannot be treated as a universal ranking of academic search services. A broader recommender survey also identifies limited user-feedback evaluation, which matters when a system is intended to help researchers explore unfamiliar areas. [Ajith et al. \(2024\)](#) [Zhang et al. \(2023\)](#)

Key papers

Scholarly recommendation systems: a literature survey. Zhang et al., 2023; 75 citations. Surveys 225 publications on scholarly recommendation, providing vocabulary and design context beyond LLM chat interfaces.

DiscipLink: Unfolding Interdisciplinary Information Seeking Process via Human-AI Co-Exploration. Zheng et al., 2024; 37 citations. Uses comparative and exploratory user studies to examine interdisciplinary search, including disciplinary terminology expansion.

LitSearch: A Retrieval Benchmark for Scientific Literature Search. Ajith et al., 2024; 64 citations. *EMNLP 2024 acceptance verified on arXiv*. Provides a reproducible retrieval benchmark; dense retrieval gains are specific to its ML and NLP tasks.

LitLLMs, LLMs for Literature Review: Are we there yet? Agarwal et al., 2025; 24 citations. *TMLR 2025; preprint first posted in 2024*. Separates retrieval, reranking, planning, and writing, clarifying where a literature-review assistant can succeed or fail.

Use and critique

Use AI-generated synonyms and citation neighborhoods to broaden a documented search. Check whether the resulting collection spans competing theories, methods, dates, languages, and publication types. Treat a missing topic as a search question, not immediately as an absence of research. This is a practical inference from the discovery and evaluation studies above.

Search terms: scholarly information retrieval; semantic search; dense retrieval; citation recommendation; citation networks; bibliographic coupling; co-citation analysis; query expansion; relevance feedback; literature discovery

```
("literature search" OR "scholarly information retrieval" OR "literature discovery") AND  
("artificial intelligence" OR "dense retrieval" OR "large language model")
```

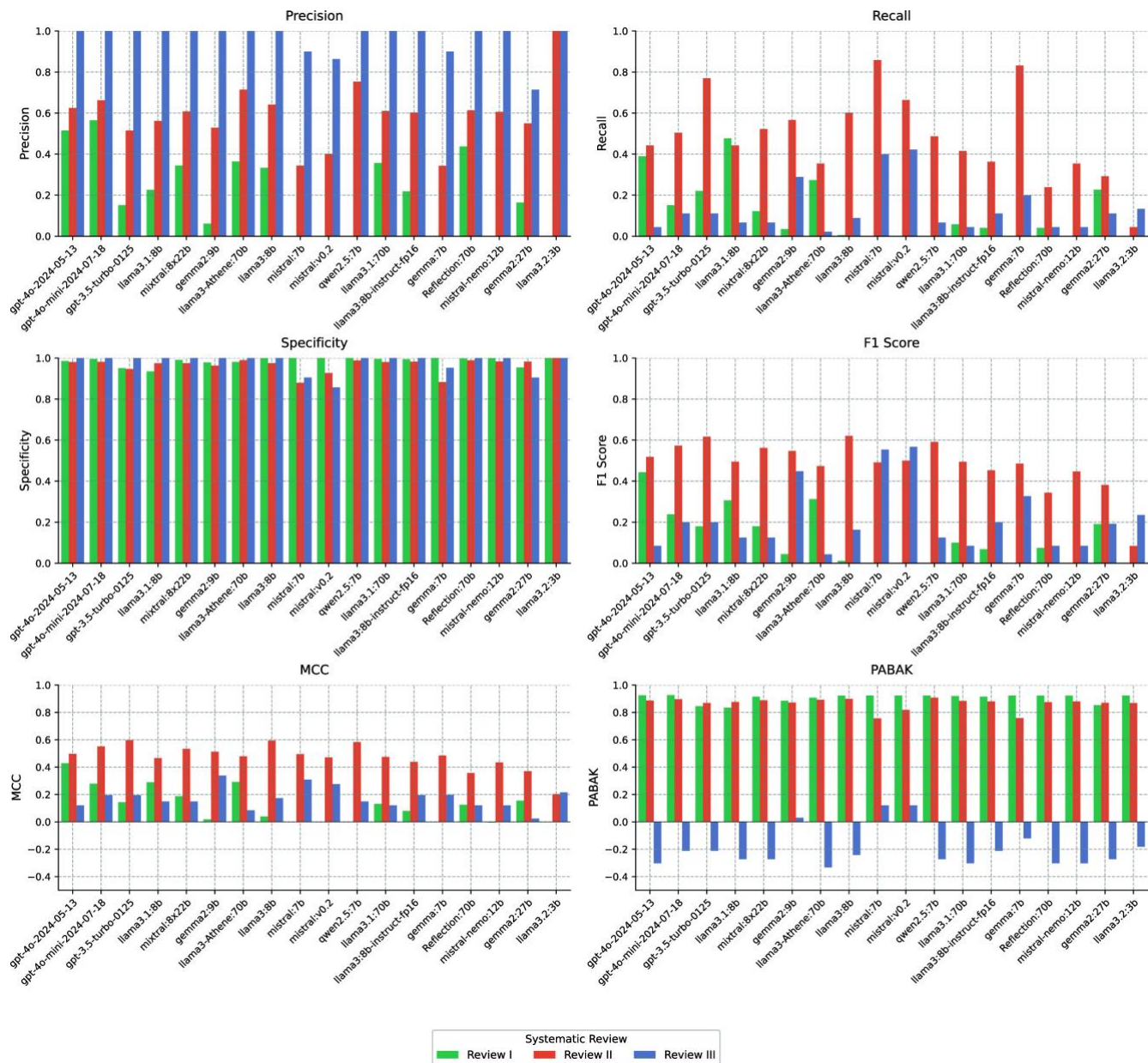
```
("citation network*" OR "bibliographic coupling" OR "co-citation" OR "research  
recommendation*") AND (discovery OR coverage OR interdisciplin* OR evaluat*)
```

2 Selecting and organizing evidence

Active learning can prioritize likely relevant records, enabling researchers to find eligible studies earlier. In education and educational psychology, a retrospective simulation across 27 reviews reported an average 58% reduction in screening workload for irrelevant records, with performance depending on the algorithm and stopping rule. That number is not a measured 58% reduction in total review-production time. [van de Schoot et al. \(2021\)](#) [Campos et al. \(2024\)](#)

LLM screening evidence is less uniform. SESR-Eval includes 34,528 labeled records from 24 software-engineering reviews; its authors found that review-to-review differences exceeded many model-to-model differences and did not recommend autonomous screening. A multilingual study likewise shows why high raw agreement can be misleading when most records are excluded. [Huotala et al. \(2025\)](#) [Khraisha et al. \(2024\)](#)

Figure 1 Screening precision and coverage are different outcomes



Compare precision in the upper-left panel with recall in the upper-right panel. High precision can coexist with low recall, meaning that many relevant records are missed. The colors represent three review datasets, illustrating why screening performance must be validated for the review at hand.

Interpretive limit. This study evaluated 18 models across three health-related systematic reviews. The figure does not establish performance across all disciplines or current model versions.

Source: [Delgado-Chaves et al. \(2025\)](#), PNAS, original Figure 1. Reproduced unchanged. [Original figure and caption](#). License: CC BY 4.0.

Key papers

An open source machine learning framework for efficient and transparent systematic reviews. van de Schoot et al., 2021; 989 citations. A highly cited foundation for interactive, open-source active-learning screening rather than autonomous review authorship.

Screening Smarter, Not Harder: A Comparative Analysis of Machine Learning Screening Algorithms and Heuristic Stopping Criteria for Systematic Reviews in Educational Research. Campos et al., 2024; 70 citations. Directly extends evaluation to educational research and tests stopping heuristics alongside screening algorithms.

SESR-Eval: Dataset for Evaluating LLMs in the Title-Abstract Screening of Systematic Reviews. Huotala et al., 2025; 10 citations. A substantial non-biomedical benchmark showing why a general claim that LLMs can replace screening is premature.

Performance of active learning models for screening prioritization in systematic reviews: a simulation study into the Average Time to Discover relevant records. Ferdinands et al., 2023; 75 citations. Compares active-learning models across six datasets and introduces Average Time to Discovery as a complementary measure.

Can large language models replace humans in systematic reviews? Evaluating GPT-4's efficacy in screening and extracting data from peer-reviewed and grey literature in multiple languages. Khraisha et al., 2024; 244 citations. *REPEAT HIT across 3 sub-areas*. Tests screening and extraction with peer-reviewed, grey, and multilingual sources; adjusts interpretation for chance agreement.

Use and critique

Evaluate recall or sensitivity alongside precision and workload. Predefine how uncertain records and stopping decisions will be handled. The SAFE procedure is a proposed practical heuristic, not a guarantee that no studies will be missed. Apply such rules in the context of the dataset and the cost of erroneous exclusion. [Boetje and van de Schoot \(2024\)](#)

Search terms: study selection; eligibility screening; title and abstract screening; full-text screening; active learning; technology-assisted review; screening prioritization; work saved over sampling; recall; stopping criteria

```
("systematic review" OR "scoping review" OR "evidence synthesis") AND (screen* OR "study selection") AND ("active learning" OR "machine learning" OR "large language model")
```

```
("AI-assisted screening" OR "technology-assisted review" OR "screening prioritization") AND (recall OR sensitivity OR "stopping criteria" OR "work saved") AND (education OR psychology OR "software engineering")
```

3 Reading extracting and appraising

Structured extraction has meaningful applications beyond medicine. Materials-chemistry work demonstrates fine-tuned entity and relation extraction, while an ecological case study reports high performance on categorical information but weaker performance on some quantitative variables. The ecological model extracted data more than 50 times faster than its human comparator, but quality assurance remains additional work. [Dagdelen et al. \(2024\)](#) [Gougherty and Clipp \(2024\)](#)

A prospective comparison within six ongoing reviews provides stronger workflow evidence: 9,341 data elements from 63 studies were assessed, and human-verified AI extraction achieved 91.0% accuracy versus 89.0% for human-only extraction, with a median 41 minutes saved per study. This supports that specific assisted workflow; it does not establish autonomous accuracy or a transferable advantage for every discipline. [Gartlehner et al. \(2025\)](#)

Table generation adds another challenge: a correct value can still be placed under the wrong heading or compared with an incompatible measure. arXiv2Table explicitly evaluates schema coverage, individual cells, and relationships across entries. Evidence for independent methodological appraisal is thinner in this selected set than evidence for extracting reported facts. [Wang et al. \(2026\)](#) [Foley et al. \(2025\)](#)

Key papers

Structured information extraction from scientific text with large language models. Dagdelen et al., 2024; 734 citations. Shows that scientific extraction includes entities and relationships in materials chemistry, not only clinical outcomes.

Testing the reliability of an AI-based large language model to extract ecological information from the scientific literature. Gougherty and Clipp, 2024; 51 citations. An ecological test illustrates why extraction performance should be broken down by information type.

Artificial Intelligence-Assisted Data Extraction With a Large Language Model: A Study Within Reviews. Gartlehner et al., 2025; 23 citations. A prospective workflow comparison measures verified accuracy and time, providing a useful model for evaluation design.

arXiv2Table: Toward Realistic Benchmarking and Evaluation for LLM-Based Literature-Review Table Generation. Wang et al., 2026; 8 citations. *ACL 2026 conference paper; index year 2025*. Tests realistic review-table construction with distractor papers and a schema that the model must develop.

Deep literature reviews: an application of fine-tuned language models to migration research. Iacus et al., 2025; 2 citations. *arXiv preprint; peer-reviewed version not established*. An arXiv application to more than 20,000 migration articles combines domain adaptation with human correction; independent transfer validation remains needed.

Use and critique

Build an extraction schema that distinguishes reported observations from reviewer interpretation. For each entry, retain a source passage or page and mark absent information explicitly. Check units, denominators, sample definitions, estimates versus uncertainty intervals, and whether a value describes the same population or time point across studies. These are practical safeguards inferred from the variable-specific and relational errors above.

Search terms: scientific information extraction; named entity recognition; relation extraction; evidence tables; literature-review table generation; data extraction; numerical fidelity; methodological appraisal; risk of bias; human verification

```
("scientific information extraction" OR "data extraction" OR "literature review table*") AND  
("large language model*" OR "generative AI") AND (accuracy OR omission* OR evaluat*)  
  
("quality appraisal" OR "critical appraisal" OR "risk of bias") AND ("artificial intelligence"  
OR "large language model*") AND (review* OR synthesis)
```

4 Synthesizing and writing

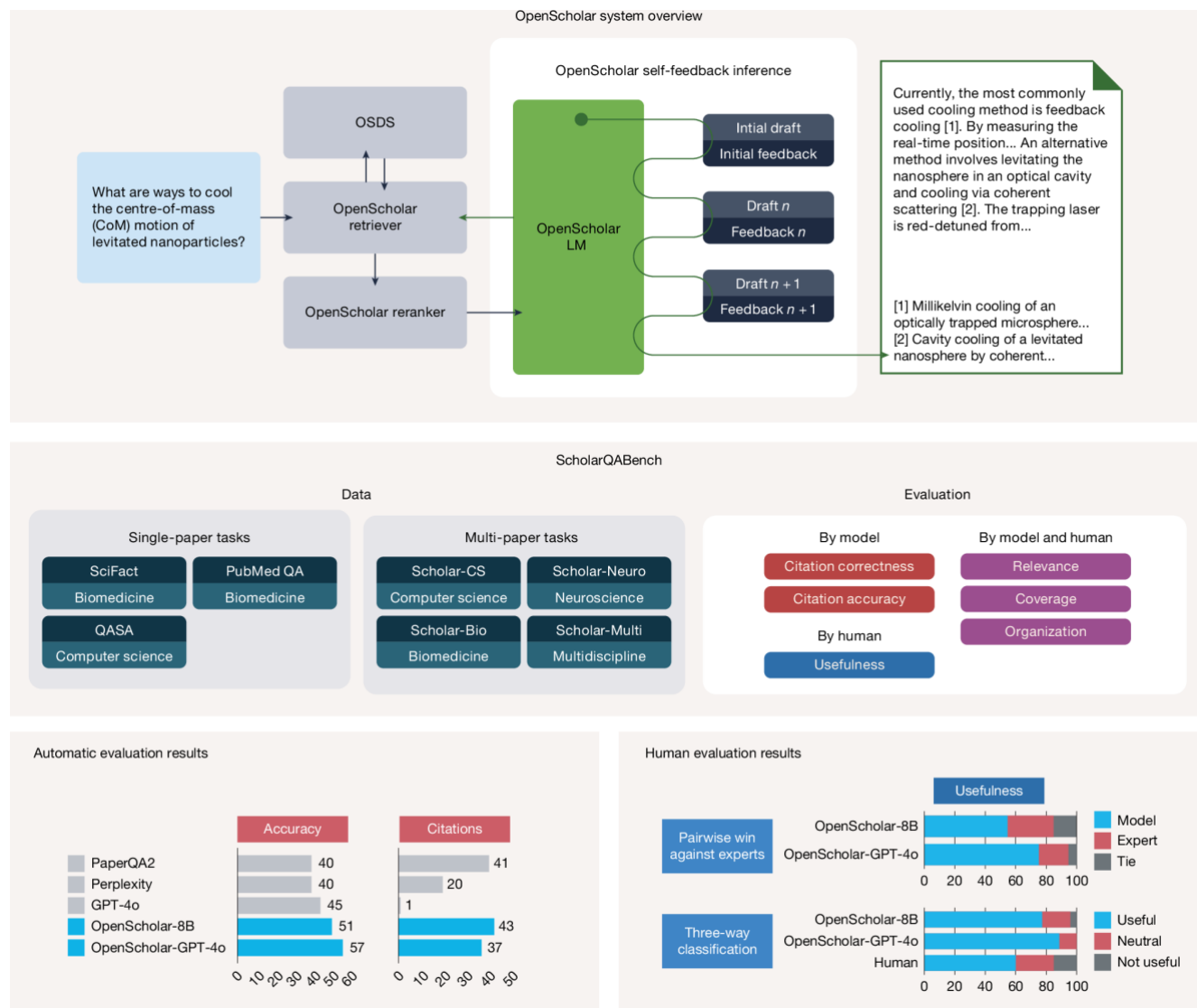
Retrieval-grounded systems can synthesize across source passages with substantially better support than an unaided model in the same evaluation. OpenScholar combines a large open-access corpus with retrieval and feedback; its benchmark spans computer science, physics, neuroscience, and biomedicine. The tested outcome is cited scientific answering, not the methodological completeness of every review genre. [Asai et al. \(2026\)](#)

Long-form evaluation remains demanding. DeepScholar-Bench reports that no evaluated system exceeded a 31% geometric mean across its metrics; this is a composite benchmark score, not 31% factual accuracy. ReportBench similarly finds room to improve coverage and factual consistency. Both are arXiv preprints and warrant reading as emerging evaluation evidence. [Patel et al. \(2025\)](#) [Li et al. \(2025\)](#)

For qualitative synthesis, interpretive assumptions and verification demands become central. Foley and colleagues offer a reflexive conceptual account; Pizard and colleagues report two trials through collaborative

autoethnography. They contribute methodological insight, but neither supplies a broad effect estimate for all qualitative review settings. [Foley et al. \(2025\)](#) [Pizard et al. \(2025\)](#)

Figure 2 Retrieval and revision support scientific synthesis



The upper panel shows how OpenScholar retrieves and reranks sources, drafts a cited answer, and revises it with feedback. The lower panels compare benchmark performance and expert evaluations. Together they illustrate how a structured pipeline can improve scientific answering.

Interpretive limit. These results concern the evaluated scientific question-answering tasks. They do not establish that AI can independently produce a complete and comprehensive literature review.

Source: [Asai et al. \(2026\)](#), Nature, original Figure 1. Reproduced unchanged. [Original figure and caption](#). License: CC BY-NC-ND 4.0. Noncommercial redistribution of the unchanged figure.

Key papers

Synthesizing scientific literature with retrieval-augmented language models. Asai et al., 2026; 78 citations.

REPEAT HIT across 4 sub-areas. A current multi-domain test of citation-grounded scientific synthesis; its repeat-hit pattern makes it a priority bridge paper.

ReportBench: Evaluating Deep Research Agents via Academic Survey Tasks. Li et al., 2025; 35 citations. *arXiv preprint; peer-reviewed version not established*. Evaluates cited literature and statement faithfulness separately; useful for understanding why a long report can still be incomplete.

DeepScholar-Bench: A Live Benchmark and Automated Evaluation for Generative Research Synthesis. Patel et al., 2025; 32 citations. *arXiv preprint; peer-reviewed version not established*. Assesses retrieval, verifiability, and knowledge synthesis jointly instead of relying on writing fluency alone.

Developing a Critical Imagination for How Researchers can use Artificially Intelligent Tools Reflexively and Responsibly During Qualitative Literature Reviews. Foley et al., 2025; 9 citations. Directly addresses qualitative literature reviewing and the relationship between AI tools and interpretive responsibility.

On the Use of Large Language Models for Qualitative Synthesis. Pizard et al., 2025; 2 citations. *2025 preprint; Consensus also lists 2026 workshop proceedings*. An exploratory account of qualitative synthesis failures and the practical effort required to verify plausible outputs.

Use and critique

Synthesize relationships among studies: agreement, disagreement, methodological differences, contextual limits, and what remains uncertain. Require support for each empirical claim. Preserve negative findings and minority explanations rather than forcing a single conclusion. Automated similarity scores and model-generated judgments require validation: one medical multi-document study found that automated rankings could disagree, even inversely, with human assessments. [Wang et al. \(2023\)](#)

Search terms: multi-document summarization; literature survey generation; related-work generation; retrieval-augmented generation; citation-grounded synthesis; thematic synthesis; qualitative evidence synthesis; conceptual integration; faithfulness; contradiction detection

```
("literature review generation" OR "scientific survey generation" OR "multi-document summarization") AND ("large language model*" OR "retrieval augmented generation") AND (faithfulness OR citation* OR coverage)
```

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("qualitative literature review*" OR "qualitative evidence synthesis" OR "thematic synthesis") AND ("generative AI" OR "large language model*") AND (reflexiv* OR interpret* OR evaluat*)
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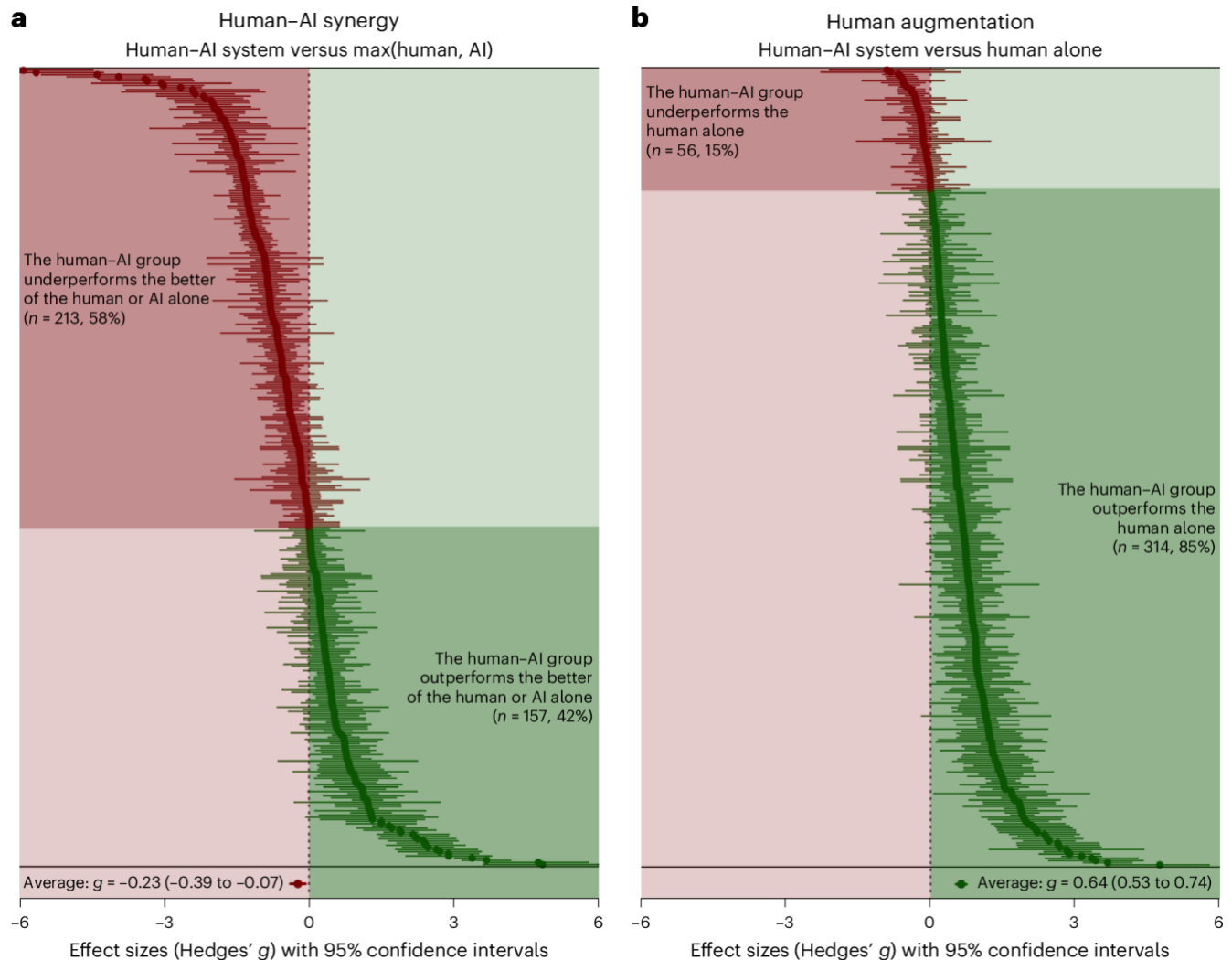
5 Reliability and human AI collaboration

Citation existence, bibliographic accuracy, claim support, and coverage are distinct outcomes. A multidisciplinary audit of 636 references across 84 generated reviews found fabricated citations in 55% of GPT-3.5 references and 18% of GPT-4 references. Even real references contained substantive errors. These are historical model-and-prompt results, not current error rates for every AI research service. [Walters and Wilder \(2023\)](#)

A second citation study in systematic-review searching reinforces the need to verify references and retrieval completeness. Conversely, the broader human–AI meta-analysis found that combined systems underperformed the better standalone comparator on average, with substantial task differences. That meta-analysis is adjacent evidence about collaboration, not a direct estimate of literature-review effectiveness. [Chelli et al. \(2024\)](#) [Vaccaro et al. \(2024\)](#)

The emerging methodological response is to specify the AI contribution, human decisions, source access, and evaluation. PRISMA-trAlce is a useful reporting proposal, but its authors explicitly describe a path toward future consensus and endorsement. It should not be presented as an already formally endorsed PRISMA extension. [Holst et al. \(2025\)](#) [Wagner et al. \(2026\)](#)

Figure 3 The comparator changes the conclusion about collaboration



The right panel compares human-AI combinations with humans alone; the left compares them with the better of humans or AI alone. The pooled effects at the bottom show improvement over humans alone but underperformance against the better standalone partner on average. The plotted observations are effect sizes, not individual participants.

Interpretive limit. This meta-analysis concerns broad human-AI collaboration, not literature-review workflows specifically. Its application here is a reason to evaluate the full workflow rather than assume that human involvement guarantees better results.

Source: [Vaccaro, Almaatouq, and Malone \(2024\)](#), Nature Human Behaviour, original Figure 1. Reproduced unchanged. [Original figure and caption](#). License: CC BY 4.0.

Key papers

Fabrication and errors in the bibliographic citations generated by ChatGPT. Walters and Wilder, 2023; 450 citations. Provides multidisciplinary evidence that fabricated citations and errors in genuine citations must be audited separately.

When combinations of humans and AI are useful: A systematic review and meta-analysis. Vaccaro et al., 2024; 533 citations. A meta-analysis of 106 experiments and 370 effect sizes shows that collaboration quality depends on task design; transfer to reviewing is an inference.

Artificial intelligence and illusions of understanding in scientific research. Messeri and Crockett, 2024; 624 citations. Provides the conceptual vocabulary of illusions of understanding and scientific monocultures without estimating their prevalence.

Generative artificial intelligence for literature reviews. Wagner et al., 2026; 3 citations. A recent information-systems perspective connects practical GenAI use with review methodology and longer-term scientific questions.

Transparent Reporting of AI in Systematic Literature Reviews: Development of the PRISMA-trAIce Checklist. Holst et al., 2025; 17 citations. Provides candidate reporting items for AI used as a review method; its proposed status is essential context.

Use and critique

Evaluate the entire workflow against a meaningful comparator. Count setup, supervision, correction, and documentation time; examine whether the final conclusions change. Record model version, access to source text, prompts, human adjudication, and any unverified outputs. A free-versus-paid comparison also needs to hold retrieval access, model capability, and task instructions constant before attributing differences to subscription status alone. The latter is an evaluation recommendation, not an established finding from this corpus.

Search terms: citation fabrication; bibliographic accuracy; attribution; citation completeness; hallucination; coverage bias; human-in-the-loop; verification burden; reproducibility; automation bias

("literature review*" OR "scientific writing") AND (ChatGPT OR "large language model*" OR "generative AI") AND ("citation fabrication" OR hallucinat* OR "bibliographic accuracy" OR "synthesis error*")

("AI-assisted" OR "human AI collaboration") AND ("literature review*" OR "evidence synthesis") AND (reproducib* OR transparen* OR "verification time" OR cost OR access)

Controlled vocabulary note [not from Consensus]: the strings above are proposed free-text starting points. For PubMed, candidate MeSH headings include Artificial Intelligence, Machine Learning, Natural Language Processing, and Information Storage and Retrieval. Verify current headings and database syntax when translating the strategy. Do not apply biomedical indexing or human-subject filters across the whole interdisciplinary search; many relevant benchmarks evaluate papers and outputs rather than participants.

Reconciling apparently conflicting findings

Apparent conflict	What differs	Interpretation
PNAS screening workload gains versus cautious software-engineering results Delgado-Chaves et al. (2025) Huotala et al. (2025)	Three reviews and 18 models versus 24 reviews and nine models; different criteria and evaluation targets.	Useful workload assistance can coexist with insufficient reliability for automatic exclusion.
Strong scientific answers versus weak long-form report scores Asai et al. (2026) Patel et al. (2025)	Different source collections, output lengths, task definitions, and scoring functions.	Do not compare the headline percentages as estimates of the same capability.
High extraction speed versus persistent numerical errors Gougherty and Clipp (2024) Gartlehner et al. (2025)	Information type and whether human verification is part of the tested workflow.	Report verified output quality and net effort, rather than generation speed alone.

Apparent conflict	What differs	Interpretation
High-quality automated chemistry reviews versus broader concerns Wu et al. (2025) Li et al. (2025)	Domain-specific pipeline with expert checks versus diverse report tasks and coverage audits.	A successful constrained application does not establish cross-disciplinary generality.
Adding a human helps in one workflow but not in an aggregate comparison Gartlehner et al. (2025) Vaccaro et al. (2024)	Task allocation, comparator, and which human contribution is measured.	Human involvement is a design variable; its value must be tested.

Across these comparisons, the defensible synthesis is conditional: effectiveness belongs to a specific combination of task, corpus, model, prompts, and human work. Results should be transferred to a new review only after examining those conditions.

Key research groups

The recurring authors below appear on at least two distinct fetched papers; repeat appearances of the same paper are not counted as independent publications. This is a map of the retrieved sample, not a global productivity ranking. Affiliations refer to the cited publication and may differ from current appointments.

Recurring author and group	Affiliation at publication	Coverage and representative work
Rens van de Schoot and Gerbrich Ferdinands	Methodology and Statistics, Utrecht University	Active learning, screening evaluation, open workflows. van de Schoot et al. (2021) Ferdinands et al. (2023)
Jonathan de Bruin	Research and Data Management Services, Utrecht University	Screening infrastructure and reproducible software; coauthor on the framework and simulation study. van de Schoot et al. (2021) Ferdinands et al. (2023)
James Thomas	EPPI-Centre, UCL Institute of Education	Evidence-synthesis methods and study identification. Ananiadou et al. (2009) O'Mara-Eves et al. (2015)
Sophia Ananiadou	National Centre for Text Mining, University of Manchester	Text mining and natural-language processing for evidence retrieval. Ananiadou et al. (2009) O'Mara-Eves et al. (2015)
Gerit Wagner with Roman Lukyanenko and Guy Paré	HEC Montréal in the 2022 publication	Information-systems review methodology, GenAI workflows, and conceptual implications. Wagner et al. (2022) Wagner et al. (2026)

The Open University team of Bolaños, Salatino, Osborne, and Motta is also a useful orientation point for tool evaluation, although its repeated appearances here mostly reflect one review rather than multiple independent publications. [Bolaños et al. \(2024\)](#)

Repeat hits and citation signals

A repeat hit means the same Consensus paper ID appeared in searches assigned to at least three distinct sub-areas. Follow-ups within one sub-area do not create additional sub-area counts. Four papers met this rule. Their breadth makes them useful connecting readings, but retrieval repetition is not proof of foundational status or independent corroboration.

Paper	Search numbers	Distinct sub-areas
Asai et al. (2026)	1, 5, 7, 16, 19	4
Khraisha et al. (2024)	4, 5, 16	3
Wu et al. (2025)	5, 12, 19	3
Wagner et al. (2022)	6, 11, 17	3

Citation counts are the Consensus snapshot retrieved for this report. The annualized indicator is citations divided by $\max(1, 2026 - \text{publication year})$; it is a rough calendar-year normalization, not a measured annual citation trajectory. Current-year papers receive a one-year denominator. Counts for separate preprint and published records are not added. Field, publication age, database coverage, and negative citations all affect interpretation.

Paper	Year used	Citations	Approximate citations per year
van de Schoot et al. (2021)	2021	989	197.8
Dagdelen et al. (2024)	2024	734	367.0
O'Mara-Eves et al. (2015)	2015	685	62.3
Messerli and Crockett (2024)	2024	624	312.0
Vaccaro et al. (2024)	2024	533	266.5
Walters and Wilder (2023)	2023	450	150.0
Wagner et al. (2022)	2022	259	64.8
Bolaños et al. (2024)	2024	233	116.5
Asai et al. (2026)	2026	78	78.0
Patel et al. (2025)	2025	32	32.0

The highest-cited directly relevant methods paper available when the citation-follow-up search was chosen was the ASReview framework. Very highly cited general reviews of LLMs were excluded from that decision because they did not directly investigate literature-review practice. Nature, Nature Machine Intelligence, Nature Communications, Nature Human Behaviour, PNAS, and field-specific methods journals appear in the selected literature. No directly relevant Science paper was established by these searches.

Open questions and gaps

Methodological gaps

Incomplete end-to-end evaluation. Many studies evaluate one stage, one benchmark, or one domain, and relatively few assess final review conclusions and total researcher effort together. Future studies should compare human-only, assisted, and more autonomous workflows using the same source access and explicit adjudication. Why it matters: local efficiency gains can disappear when verification and correction are included. [Bolaños et al. \(2024\)](#) [Gartlehner et al. \(2025\)](#) [Wagner et al. \(2026\)](#)

Unstable or incomplete reference standards. Published reference lists are useful exemplars, but an exact-match target can penalize a relevant alternative source or reward a familiar omission. LLM judges and automated summarization metrics introduce their own error. Why it matters: improving a benchmark score may not improve scholarly adequacy. This is a methodological inference from the design of [Agarwal et al. \(2025\)](#), [Li et al. \(2025\)](#), and [Wang et al. \(2023\)](#).

Limited replication and task-sensitive results. Model versions, prompts, class imbalance, stopping rules, and source availability vary across studies. Large record counts do not necessarily mean many independent review topics or researchers were tested. Why it matters: tens of thousands of abstracts can still provide a narrow basis for claims about review practice. [O'Mara-Eves et al. \(2015\)](#) [Huotala et al. \(2025\)](#) [Khraisha et al. \(2024\)](#)

Population and context gaps

Uneven disciplinary and language coverage. The retrieved evidence includes meaningful non-biomedical studies, but many benchmarks remain concentrated in computational or biomedical literature. Humanities scholarship, books, archival sources, critical and realist reviews, and many non-English traditions are not well validated in this set. Why it matters: differences in publication form and epistemology can change both retrieval and synthesis quality. [Ajith et al. \(2024\)](#) [Asai et al. \(2026\)](#) [Foley et al. \(2025\)](#)

Limited evidence on researcher experience and access. The corpus does not establish how well findings transfer across novice and expert reviewers, institutions with different subscriptions, or researchers using free versus paid versions. Why it matters: a workflow may save effort for a well-resourced specialist while increasing verification or access burdens for someone else. The cost-and-equity questions remain open here rather than demonstrated outcomes. [Bolaños et al. \(2024\)](#) [Wagner et al. \(2026\)](#)

Sparse sustained field use. Benchmarks, retrospective simulations, exploratory demonstrations, and small qualitative trials answer different questions from repeated use across an actual review project. Why it matters: researcher learning, fatigue, changing questions, and model updates can alter effectiveness over time. [Campos et al. \(2024\)](#) [Zheng et al. \(2024\)](#) [Pizard et al. \(2025\)](#)

Conceptual and theoretical gaps

No shared definition of successful synthesis. Citation accuracy, factual correctness, retrieval breadth, thematic richness, theory development, and reader preference are related but distinct. Why it matters: a system may excel at one and fail at another, making broad claims that it can “do a literature review” difficult to assess. [Wagner et al. \(2022\)](#) [Patel et al. \(2025\)](#) [Foley et al. \(2025\)](#)

Unresolved effects on understanding and intellectual diversity. AI may help researchers cross disciplinary boundaries while also reinforcing familiar questions and perspectives. Why it matters: improved access to papers need not produce broader or more critical understanding. The relationship between these mechanisms needs direct testing. [Zheng et al. \(2024\)](#) [Messerli and Crockett \(2024\)](#)

Human oversight lacks a sufficiently precise theory of task allocation. Knowing that a researcher remains involved says little about what they inspect, when they intervene, or how they detect a convincing error. Why it matters: effective collaboration requires more than assigning nominal responsibility to a human. [Vaccaro et al. \(2024\)](#) [Pizard et al. \(2025\)](#)

Practical implications for using and critiquing AI

The following workflow is a synthesis of the evidence rather than a separately validated protocol. It is suitable as a planning aid for research training and should be adapted to the review genre.

Stage	Useful AI contribution	Researcher judgment to retain
Frame	Generate candidate questions, synonyms, and a preliminary concept map.	Set the purpose, review genre, theoretical perspective, and boundaries.
Find	Expand terminology, retrieve candidates, and follow citation connections.	Evaluate coverage and search beyond the tool's visible source collection.
Select	Rank records and surface reasons for relevance.	Validate eligibility decisions, uncertainty handling, and stopping rules.
Extract	Populate a structured matrix with linked source passages.	Verify numbers, context, missing information, and methodological judgments.
Synthesize	Draft comparisons, identify apparent tensions, and organize themes.	Determine whether differences are substantive, methodological, contextual, or unresolved.
Audit	Flag unsupported statements, inconsistent entries, and citation problems.	Check the final argument against original sources and document the actual AI contribution.

For a class repository, a balanced pairing would place a capability study next to a critique using a different outcome: DiscipLink with the qualitative reflexivity paper; the education screening study with SESR-Eval; or OpenScholar with DeepScholar-Bench. Such pairings allow discussion of what was demonstrated, under which conditions, and which review responsibilities remain untested. [Zheng et al. \(2024\)](#) [Foley et al. \(2025\)](#) [Campos et al. \(2024\)](#) [Huotala et al. \(2025\)](#) [Asai et al. \(2026\)](#) [Patel et al. \(2025\)](#)

Bibliography

Alphabetical by first-author surname. Every listed paper was returned by Consensus and is linked to its complete Consensus record. Years follow the identified publication version where verified; unresolved version differences are labeled. Author lists below preserve the available record, with long lists abbreviated after the first three authors. Citation counts are snapshot metadata rather than quality ratings.

Shubham Agarwal, Gaurav Sahu, Abhay Puri, et al. (2025). [LitLLMs, LLMs for Literature Review: Are we there yet?](#) Trans. Mach. Learn. Res. DOI: 10.48550/arxiv.2412.15249. Consensus citations: 24. *TMLR 2025; preprint first posted in 2024.*

Anirudh Ajith, Mengzhou Xia, Alexis Chevalier, Tanya Goyal, Danqi Chen, Tianyu Gao (2024). [LitSearch: A Retrieval Benchmark for Scientific Literature Search](#). EMNLP 2024; arXiv version returned by Consensus. DOI: 10.48550/arxiv.2407.18940. Consensus citations: 64. *EMNLP 2024 acceptance verified on arXiv.*

S. Ananiadou, Brian Rea, Naoaki Okazaki, R. Procter, James Thomas (2009). [Supporting Systematic Reviews Using Text Mining](#). Social Science Computer Review. DOI: 10.1177/0894439309332293. Consensus citations: 142.

Akari Asai, Jacqueline He, Rulin Shao, et al. (2026). [Synthesizing scientific literature with retrieval-augmented language models](#). Nature. DOI: 10.1038/s41586-025-10072-4. Consensus citations: 78.

Josien Boetje, Rens van de Schoot (2024). [The SAFE procedure: a practical stopping heuristic for active learning-based screening in systematic reviews and meta-analyses](#). Systematic Reviews. DOI: 10.1186/s13643-024-02502-7. Consensus citations: 120.

Francisco Bolanos, Angelo Salatino, Francesco Osborne, Enrico Motta (2024). [Artificial intelligence for literature reviews: opportunities and challenges](#). Artificial Intelligence Review. DOI: 10.1007/s10462-024-10902-3. Consensus citations: 233.

Diego G. Campos, R. Lavelle-Hill, Kou Murayama, Martin Hecht, Steffen Zitzmann, Ronny Scherer (2024). [Screening Smarter, Not Harder: A Comparative Analysis of Machine Learning Screening Algorithms and Heuristic Stopping Criteria for Systematic Reviews in Educational Research](#). Educational Psychology Review. DOI: 10.1007/s10648-024-09862-5. Consensus citations: 70.

Michaël Chelli, J. Descamps, V. Lavoué, et al. (2024). [Hallucination Rates and Reference Accuracy of ChatGPT and Bard for Systematic Reviews: Comparative Analysis](#). Journal of Medical Internet Research. DOI: 10.2196/53164. Consensus citations: 406.

John Dagdelen, Alex Dunn, Sanghoon Lee, et al. (2024). [Structured information extraction from scientific text with large language models](#). Nature Communications. DOI: 10.1038/s41467-024-45563-x. Consensus citations: 734.

José de la Torre-López, Aurora Ramírez, José Raúl Romero (2023). [Artificial intelligence to automate the systematic review of scientific literature](#). Computing. DOI: 10.1007/s00607-023-01181-x. Consensus citations: 158.

F. Delgado-Chaves, Matthew J. Jennings, A. Atalaia, et al. (2025). [Transforming literature screening: The emerging role of large language models in systematic reviews](#). Proceedings of the National Academy of Sciences of the United States of America. DOI: 10.1073/pnas.2411962122. Consensus citations: 84.

Gerbrich Ferdinands, Raoul Schram, Jonathan de Bruin, et al. (2023). [Performance of active learning models for screening prioritization in systematic reviews: a simulation study into the Average Time to Discover relevant records](#). Systematic Reviews. DOI: 10.1186/s13643-023-02257-7. Consensus citations: 75.

Kristen Foley, C. McLean, Reece De Zylva, et al. (2025). [Developing a Critical Imagination for How Researchers can use Artificially Intelligent Tools Reflexively and Responsibly During Qualitative Literature Reviews](#). International Journal of Qualitative Methods. DOI: 10.1177/16094069251316249. Consensus citations: 9.

Gerald Gartlehner, Shannon Kugley, Karen Crotty, et al. (2025). [Artificial Intelligence-Assisted Data Extraction With a Large Language Model: A Study Within Reviews](#). Annals of internal medicine. DOI: 10.7326/annals-25-00739. Consensus citations: 23.

Andrew V. Gougherty, H. Clipp (2024). [Testing the reliability of an AI-based large language model to extract ecological information from the scientific literature](#). npj Biodiversity. DOI: 10.1038/s44185-024-00043-9. Consensus citations: 51.

Dirk Holst, Keno Moenck, Julian Koch, Ole Schmedemann, Thorsten Schüppstuhl (2025). [Transparent Reporting of AI in Systematic Literature Reviews: Development of the PRISMA-trAlce Checklist](#). JMIR AI. DOI: 10.2196/80247. Consensus citations: 17.

Aleksi Huotala, Miikka Kuuttila, Mika Mäntylä (2025). [SESR-Eval: Dataset for Evaluating LLMs in the Title-Abstract Screening of Systematic Reviews](#). 2025 ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM). DOI: 10.1109/esem64174.2025.00053. Consensus citations: 10.

S. Iacus, Haodong Qi, Jiyoung Han (2025). [Deep literature reviews: an application of fine-tuned language models to migration research](#). ArXiv. DOI: 10.48550/arxiv.2504.13685. Consensus citations: 2. *arXiv preprint; peer-reviewed version not established*.

Qusai Khraisha, Sophie Put, Johanna Kappenberg, Azza Warraitch, K. Hadfield (2024). [Can large language models replace humans in systematic reviews? Evaluating GPT-4's efficacy in screening and extracting data from peer-reviewed and grey literature in multiple languages](#). Research Synthesis Methods. DOI: 10.1002/jrsm.1715. Consensus citations: 244. *Journal year 2024; Consensus indexes 2023*.

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Lisa Messeri, M. J. Crockett (2024). [Artificial intelligence and illusions of understanding in scientific research](#). Nature. DOI: 10.1038/s41586-024-07146-0. Consensus citations: 624.

A. O'Mara-Eves, James Thomas, J. McNaught, Makoto Miwa, S. Ananiadou (2015). [Using text mining for study identification in systematic reviews: a systematic review of current approaches](#). Systematic Reviews. DOI: 10.1186/2046-4053-4-5. Consensus citations: 685.

Liana Patel, Negar Arabzadeh, Harshit Gupta, et al. (2025). [DeepScholar-Bench: A Live Benchmark and Automated Evaluation for Generative Research Synthesis](#). ArXiv. DOI: 10.48550/arxiv.2508.20033. Consensus citations: 32. *arXiv preprint; peer-reviewed version not established*.

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Rens van de Schoot, Jonathan de Bruin, Raoul Schram, et al. (2021). [An open source machine learning framework for efficient and transparent systematic reviews](#). *Nature Machine Intelligence*. DOI: 10.1038/s42256-020-00287-7. Consensus citations: 989. *Journal year 2021; Consensus indexes 2020*.

Gerit Wagner, R. Lukyanenko, G. Paré (2022). [Artificial intelligence and the conduct of literature reviews](#). *Journal of Information Technology*. DOI: 10.1177/02683962211048201. Consensus citations: 259. *First published online in 2021; journal issue 2022*.

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Appendix on search and metadata

Search log

Search date: 15 September 2026. One broad reconnaissance query preceded the 20-query expansion below. Searches were sequential. Year ranges were applied to the historical and contemporary screening searches. No human-subject, sample-size, or journal-quartile filter was imposed; those restrictions would exclude relevant computational work, and sample-size and SJR filters were not documented by the connector. Results were capped by the returned result set, so the inventory does not measure total database coverage.

Reconnaissance query: Artificial intelligence machine learning large language models for literature reviews evidence synthesis search screening data extraction hallucinated citations coverage accuracy systematic reviews. Returned 20 records.

01 Sub-area 1 | 20 returned records

AI scholarly information retrieval semantic search citation recommendation literature discovery evaluation interdisciplinary research

02 Sub-area 1 | 20 returned records

artificial intelligence literature review bibliometric science mapping knowledge graphs social sciences humanities education

Largely off-topic: reviews about AI in education.

03 Sub-area 2 | 20 returned records

machine learning active learning literature screening systematic evidence review social science education environmental science

04 Sub-area 2 | 20 returned records

large language models paper screening benchmark systematic literature reviews software engineering computer science selection inclusion criteria

05 Sub-area 3 | 20 returned records

large language models scientific information extraction literature reviews extracting methods findings tables evaluation benchmark

06 Sub-area 3 | 20 returned records

AI assisted critical appraisal research papers methodological quality assessment literature reviews qualitative quantitative

07 Sub-area 4 | 20 returned records

automated literature survey generation multi document scientific summarization retrieval augmented generation benchmark citation faithfulness

08 Sub-area 4 | 20 returned records

large language models qualitative literature review thematic synthesis conceptual synthesis humanities social sciences

09 Sub-area 5 | 20 returned records

human AI collaboration literature review researchers user study comparison efficiency sensemaking interdisciplinary

10 Sub-area 5 | 20 returned records

ChatGPT fabricated citations bibliographic references hallucinations accuracy scientific literature reviews disciplines

11 Sub-area 5 | 20 returned records

"artificial intelligence" "literature reviews" opportunities challenges systematic review automation research methods

12 Sub-area 4 | 20 returned records

survey review automatic literature review generation scientific multi document summarization evaluation faithfulness

13 Sub-area 2 | 20 returned records

text mining machine learning automatic systematic literature review screening evidence synthesis year:1900-2015

14 Sub-area 2 | 20 returned records

active learning automated screening systematic literature reviews education stopping criteria benchmark year:2021-2026

15 Sub-area 2 | 20 returned records

"An open source machine learning framework for efficient and transparent systematic reviews" ASReview van de Schoot

16 Sub-area 4 | 20 returned records

arXiv literature search review generation evaluation benchmark LitSearch ReportBench ScholarQABench AutoSurvey

17 Sub-area 4 | 20 returned records

"qualitative literature reviews" artificial intelligence reflexivity synthesis

18 Sub-area 1 | 1 returned record

AI assisted scholarly literature search coverage bias disciplines evaluation "DiscipLink" "LitSearch"
Only one result, off-topic; no directly usable evidence.

19 Sub-area 5 | 20 returned records

AI literature review synthesis science Nature hallucination illusions understanding scientific research evaluation

20 Sub-area 5 | 20 returned records

"literature review" "AI" "user study" researcher writing synthesis evaluation

Version and metadata checks

The index sometimes merges an early posting year with a later journal or proceedings record. ASReview was normalized to 2021; Wagner and colleagues to the 2022 journal issue with 2021 online-first noted; Khraisha and colleagues to 2024; LitLLMs to TMLR 2025; arXiv2Table to ACL 2026; and Wu and colleagues to the 2025 journal publication. Pizard and colleagues are retained as a 2025 preprint with a separate note that the returned record lists 2026 workshop proceedings. OpenScholar's 2024 preprint and 2026 Nature article were treated as one research contribution, using the latter record; citation counts were not combined. Publisher records also corrected incomplete authorship for de la Torre-López and colleagues and for Boetje and van de Schoot, and corrected the Bagheri attribution in Ferdinands and colleagues.

LitSearch was returned as arXiv, but its arXiv record states EMNLP 2024 acceptance. Thus arXiv availability was not treated as evidence that a work lacks peer review. ReportBench, DeepScholar-Bench, and the migration application are labeled preprints because a peer-reviewed version was not established in the inspected records. DOI and venue checks do not substitute for full-text methodological review.

[Bolaños and colleagues metadata and author affiliations](#)

[O'Mara-Eves and colleagues author affiliations](#)

[Wagner and colleagues journal year and affiliations](#)

[Khraisha and colleagues publication year and methods](#)

[Ferdinands and colleagues affiliations and ASReview reference year](#)

[LitSearch publication status](#)

[LitLLMs publication status on the authors' project page](#)

[arXiv2Table publication year and proceedings](#)

[Pizard and colleagues preprint record](#)

[De la Torre-López and colleagues authorship](#)

[Boetje and van de Schoot authorship](#)

Citation repetition and annualized counts were calculated from the retrieval inventory. The report's judgments of relative evidence maturity are qualitative assessments of the selected designs, domains, and outcomes, not formal risk-of-bias grades or pooled effect estimates. The search's strong computational and biomedical representation, off-topic retrievals, and selective full-text access limit any claim to complete disciplinary coverage.